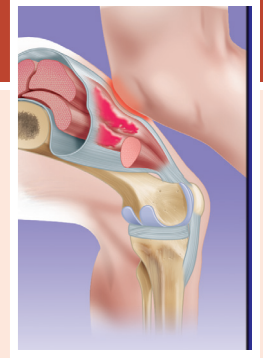


# Modalities and Manual Techniques in Sports Medicine Rehabilitation

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## MODALITIES AND MANUAL THERAPY

Modalities, physical agents, and manual therapies (MTs) are commonly applied interventions for sports-related injuries and are designed to facilitate the rehabilitation process. These treatments are often associated with pain modulation techniques that can allow an earlier implementation of therapeutic exercise. Modalities such as ice and electrical stimulation can often reduce pain, and MT techniques similarly modulate the neuromuscular system so that the focus can remain on restoring function. New methods and devices are continually marketed and purported to alleviate symptoms and treat inflammatory conditions. Clinicians must stay vigilant to address the patient's impairments and then design and implement treatments that meet specific treatment goals. Using evidence, applying the treatments properly, and measuring results are imperative throughout the process.

### Modalities

Modalities such as cryotherapy (ice), electrical stimulation, heat, ultrasound, and laser are commonly applied to the body to affect the inflammatory cascade and to reduce pain. Ice is probably the most commonly applied treatment and is part of the standard of care. For most acute injuries the acronym POLICE (*Protect, Optimal loading, Ice, Compression, Elevation*) should be followed (Fig. 32.1).<sup>1</sup> Electrical stimulation has been used in rehabilitation for thousands of years and dates back to the ancient Olympic Games in Greece when clinicians used electrical eels to treat athletes' injuries. Today electrical currents are applied as either transcutaneous electrical nerve stimulation (TENS) or as neuromuscular electrical nerve stimulation (NMES) to reduce pain or stimulate a muscle contraction. Newer forms of electrotherapy, such as patterned electrical nerve stimulation (PENS), have additional application in muscle activation and recruitment. Ultrasound and laser are often termed biostimulators since they use mechanical (acoustical energy) or light energy to stimulate cellular processes within the tissues. Thus the types of energies and equipment available for clinical use are continually expanding.

Literature that supports the use of therapeutic modalities varies and researchers often try to identify physiologic mechanisms or responses when these agents are applied. Studies on healthy individuals, as is often the case with cryotherapy studies, confound the literature since the neuromuscular changes induced with ice application would be different without pathology. Garnering

evidence to support the use of modalities is difficult combined with the fact that clinical trials are difficult to control. For example, to study whether TENS enhances pain modulation in a specific injury, the researcher would need to control all other therapies including ice, oral analgesics, and activity. Typically, once a patient feels improvement, he or she increases their activity level, which can, in turn, result in more discomfort. The use of patient-reported outcome scales, physical measures of edema, range of motion (ROM), strength, and data from activity monitors can help provide evidence for specific treatments. Furthermore, modalities all have parameters that affect the dosage, which should be consistently applied and modified when appropriate. For example, electrical currents have a variety of waveforms, phase durations, pulse frequencies, and recommended intensities to elicit the desired response. Ultrasound can vary in frequency, duty cycle, and intensity; and lasers vary in wavelength, intensity, and possibly duty cycle. The area treated, the amount of tissues exposed, and how often the treatment is delivered can also affect the outcome. Thus the clinician is expected to understand the parameters and adjust them properly according to the desired outcome. Since modalities often apply thermal, electrical, acoustical, or light energy to the body, precautions include cardiac pacemakers (or implanted electromagnetic devices), sensory loss (especially to temperature changes), and peripheral artery disease that affects normal physiology in the extremities. Specific precautions should always be applied regarding the specific technique.

### Manual Therapy

Similar to modalities, MT is a passive, nonsurgical type of conservative management that involves skilled movements applied by clinicians to the patient's body that directly or indirectly targets a variety of anatomical structures or systems.<sup>2</sup> MT is used to assess, diagnose, and treat a variety of symptoms and conditions that are intended to modulate pain; improve tissue extensibility; increase ROM; induce relaxation; mobilize or manipulate soft tissue and joints; and reduce soft tissue swelling, inflammation, or restriction.<sup>2</sup> Health care practitioners implement MT when the examination findings, diagnosis, and prognosis indicate the use of these techniques to decrease edema, pain, spasm, or swelling; enhance health, wellness, and fitness; improve or maintain physical performance; increase the ability to move; or prevent or remediate impairment in body functions and structures, activity limitations, or participation restrictions to improve physical

**Abstract**

Modalities, physical agents, and manual therapies (MTs) are commonly applied interventions for sports-related injuries and are designed to facilitate the rehabilitation process. These treatments are often associated with pain modulation techniques that can allow an earlier implementation of therapeutic exercise. Modalities such as ice and electrical stimulation can often reduce pain, and MT techniques similarly modulate the neuromuscular system so that the focus can remain on restoring function. New methods and devices are continually marketed and purported to alleviate symptoms and treat inflammatory conditions. Clinicians must stay vigilant to address the patient's impairments and then design and implement treatments that meet specific treatment goals. Using evidence, applying the treatments properly, and measuring results are imperative throughout the process.

**Keywords**

therapeutic modalities  
manual therapy  
impairment-based approach  
mobilization  
multimodal approach

|           |                 |                                                           |
|-----------|-----------------|-----------------------------------------------------------|
| <b>P</b>  | Protect         | Minimizes stress to the injured structure                 |
| <b>OL</b> | Optimal loading | Soft tissues respond well to graded and controlled stress |
| <b>I</b>  | Ice             | Reduces local metabolism & pain                           |
| <b>C</b>  | Compression     | Reduces and controls edema                                |
| <b>E</b>  | Elevation       | Reduces and controls edema                                |

**Fig. 32.1** POLICE to treat acute injuries. (From Bleakley CM, Glasgow P, MacAuley DC. PRICE needs updating, should we call the POLICE? *Br J Sports Med.* 2012;46[4]:220–221. doi:10.1136/bjsports-2011-090297.)

function. Precautions, and contraindications for MT include joint pain with the technique, joint effusion, unknown pathology, auto-immune diseases, fracture, tumor, infection and osteoporosis. MT should be avoided with nerve and vascular pathologies where movement and pressure can exacerbate the condition. Although these techniques can be helpful and useful, there is no evidence that any one of them is the only way or even the best way to treat a particular condition.

There are many MT techniques with different names and uses. There are even types that offer practitioners certifications in a particular technique. You may see these listed as trigger point release (TPR), proprioceptive neuromuscular facilitation (PNF), muscle energy technique (MET), strain-counterstrain, active release technique (ART), cranio-sacral therapy, myofascial release, positional release therapy, and others. Most of these MT techniques may be categorized into four major groups: (1) manipulation (high velocity low amplitude—HVLA), (2) mobilization (nonthrust manipulation), (3) stretching, and (4) muscle-modifying techniques.

Collectively, the process of interventions with either modalities or MT is grounded on clinical reasoning to enhance patient management for musculoskeletal pain by influencing factors from a multidimensional perspective that have potential to positively impact clinical outcomes. The influence of biomechanical, neurophysiologic, psychological, and nonspecific patient factors as treatment mediators and/or moderators provides additional information related to the process and potential mechanisms by which the treatment may be effective. Additionally, the definition and purpose of MT varies across health care professionals. As health care delivery advances toward personalized approaches, there is a crucial need to advance our understanding of the underlying mechanisms associated with MT and modality effectiveness.

Modalities have been shown to be effective when applied in a variety of ways to various pathologies. MT has been shown to be effective for increasing function and pain reduction in the treatment of musculoskeletal disorders including low back pain,<sup>3,4</sup> carpal tunnel syndrome,<sup>5,6</sup> knee osteoarthritis,<sup>7</sup> and hip osteoarthritis.<sup>8</sup> Moreover, recent studies have provided even stronger evidence when participants are classified into sub-groups.<sup>4,9</sup> Refer to Table 32.1 for a summary. Despite the literature supporting its effectiveness, the mechanisms of these treatments are not well established.

## IMPAIRMENT-BASED REHABILITATION PROGRAMS

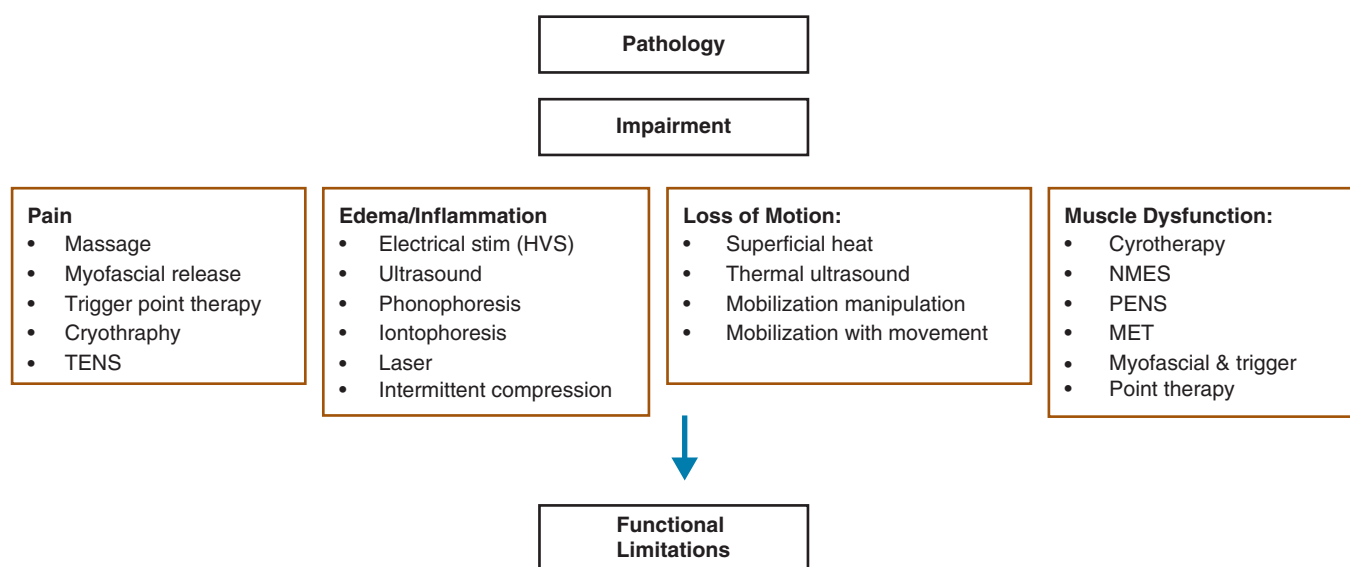
Evidence-based practice describes a clinical paradigm that prescribes certain interventions when there is evidence, based on

either research or empiricism, to support that intervention. Each treatment, whether it is a modality or a MT, should be applied with a specific goal in mind. A systematic evaluation of the injury, including biomechanical assessment should be conducted. That evaluation will reveal the presence of any impairment. For example, when evaluating an acute ankle injury, the clinician may find edema, localized pain on the lateral ankle, limited ROM, and pain with weight bearing. Those findings become the impairment list and only treatments with evidence to resolve a specific impairment should be administered (Fig. 32.2).<sup>45</sup> Evidence-based practice describes a clinical paradigm that prescribes certain interventions when there is evidence, based on either research or empiricism, to support that intervention. Each treatment, whether that is a modality or MT, should be applied with a specific goal in mind. A systematic evaluation of the injury, including biomechanical assessment should be conducted. That evaluation will reveal the presence of any impairment. For example, when evaluating an acute ankle injury, the clinician may find edema, localized pain on the lateral ankle, limited ROM, and pain with weight bearing. Those findings become the impairment list and only treatments with evidence to resolve a specific impairment should be administered. Clinicians who measure outcomes will develop their own evidence to support the continued use of an intervention or have the justification to cease the treatment. Thus the treatments and the rehabilitation program continually evolve. This chapter is not inclusive of all therapies that can or should be applied to facilitate the rehabilitation program. However, we aim to present the information in a style that might help direct a clinician to an appropriate therapy based on the results of the clinical evaluation and treatment goals.

Sports and orthopaedic injuries and pathologies cannot be resolved by these interventions alone. Time, resolution of the inflammation, and healing must occur and should be within the context of surgical repairs or reconstructions and immobilization. Tissues must be protected as they heal and the goal of both postoperative and conservative management is to restore or replace neuromuscular pathways and to provide strength and dynamic stability. Many chronic pain syndromes develop altered neural pathways that result in the inhibition of muscle function, decreased strength, and restriction of movement. Different treatments should be applied for those impairments. Sports medicine is all about returning the patient to function as quickly as possible. If an intervention might hasten the healing process by allowing earlier exercise, increasing ROM, or restoring more confidence while performing functional exercise, then best practice would suggest that the therapy should be used. Most of these treatments are safe, but keeping therapies aligned with the impairment contains costs and prevents unnecessary treatments.

TABLE 32.1 Effectiveness of Manual Therapy for Body Region

| Region               | Effectiveness of Manual Therapy (Associated With Improvements in Pain, Function, and Disability in Individuals With the Following Conditions)                                                                                                                                                                                | Effectiveness of Modalities                                                                                                                         |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Shoulder             | <ul style="list-style-type: none"> <li>Shoulder impingement syndrome<sup>10</sup></li> <li>Adhesive capsulitis<sup>11</sup></li> </ul>                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Calcific tendonitis<sup>12</sup></li> </ul>                                                                  |
| Elbow                | <ul style="list-style-type: none"> <li>Lateral epicondylalgia<sup>13,14</sup></li> </ul>                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Lateral epicondylalgia<sup>15</sup></li> </ul>                                                               |
| Wrist                | <ul style="list-style-type: none"> <li>Lateral epicondylalgia<sup>16</sup></li> <li>Carpal tunnel syndrome<sup>17</sup></li> </ul>                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>Carpal tunnel<sup>18</sup></li> </ul>                                                                        |
| Hip                  | <ul style="list-style-type: none"> <li>Hip osteoarthritis<sup>19</sup></li> </ul>                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Osteoarthritis<sup>20</sup></li> <li>Hip arthroplasty<sup>21</sup></li> <li>Hip pain<sup>22</sup></li> </ul> |
| Knee                 | <ul style="list-style-type: none"> <li>Knee osteoarthritis<sup>19</sup></li> <li>Limited evidence for PFP<sup>23</sup></li> </ul>                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Knee osteoarthritis,<sup>24</sup> patellofemoral pain<sup>25</sup></li> </ul>                                |
| Ankle <sup>107</sup> | <ul style="list-style-type: none"> <li>Ankle sprains<sup>28,29</sup></li> <li>Plantar heel pain<sup>30</sup></li> <li>Cuboid syndrome<sup>31</sup></li> <li>Symptomatic hallux abducto valgus<sup>32</sup></li> </ul>                                                                                                        | <ul style="list-style-type: none"> <li>ACL reconstruction<sup>26,27</sup></li> <li>Ankle sprains<sup>1,33</sup></li> </ul>                          |
| Cervical spine       | <ul style="list-style-type: none"> <li>Mechanical neck pain<sup>34</sup></li> <li>Cervical radiculopathy<sup>35</sup></li> <li>Whiplash associated disorder<sup>36</sup></li> <li>Cervicogenic headache<sup>36</sup></li> <li>Nonspecific shoulder pain<sup>37</sup></li> <li>Lateral epicondylalgia<sup>38</sup></li> </ul> | <ul style="list-style-type: none"> <li>Nonspecific neck &amp; trigger point pain<sup>39</sup></li> </ul>                                            |
| Thoracic spine       | <ul style="list-style-type: none"> <li>Mechanical neck pain<sup>40</sup></li> <li>Cervical radiculopathy<sup>35,40</sup></li> <li>Shoulder pain<sup>37</sup></li> </ul>                                                                                                                                                      |                                                                                                                                                     |
| Lumbar spine         | <ul style="list-style-type: none"> <li>Nonspecific low back pain<sup>41,42</sup></li> <li>Lumbar spinal stenosis<sup>108</sup></li> <li>Lumbosacral radiculopathy<sup>43</sup></li> </ul>                                                                                                                                    | <ul style="list-style-type: none"> <li>Low back pain<sup>44</sup></li> </ul>                                                                        |



**Fig. 32.2** Impairments and treatments in sports medicine conditions. (Modified from Vela and Denegar.) (From Iverson CA, Sutlive TG, Crowell MS, et al. Lumbopelvic manipulation for the treatment of patients with patellofemoral pain syndrome: development of a clinical prediction rule. *Orthop Sports Phys Ther.* 2008;38[6]:297–309; discussion 309–312. doi: 10.2519/jospt.2008.2669.)

## Pain Modulation

Pain is perhaps the most common factor in any sports injury or orthopaedic condition and is involved with the inflammatory process. Pain is useful since it alerts the patient that there is a problem, but it can begin a cascade of factors that can change motor function that is designed to protect the injured body part. For example, friction in the glenohumeral joint of a swimmer may begin with some pain. Unknown to the swimmer, the supraspinatus muscle becomes inhibited. Although there is discomfort, the swimmer continues his workout and the poor motor function prevents adequate stabilization and depression of the humeral head. Minor adjustments in the scapular position permit another 20,000 to 30,000 strokes with no real change in the ability to perform. However, the change in mechanics may result in bursitis, tendinopathy, and impingement. Resolving the pain in this case is imperative to restoring the mechanical function of the supraspinatus.

Pain is a complex neurophysiologic phenomenon and methods to treat pain include medications as well as physical agents and MT. Medications typically act on the chemical cascade of the inflammatory process, and include non-steroidal antiinflammatory drugs (NSAIDs) and steroids. Various other analgesics, both narcotic and nonnarcotic, act on the chemical neurotransmitters throughout the nervous system. Physical agents and manual therapies act by stimulating various sensory systems to elicit endogenous neurotransmitters or by creating alternative sensory input that reduces the painful input. Regardless of the method used for analgesia, it is important to restore normal stimulus to the muscle system. Returning to the swimmer, when pain is reduced, the supraspinatus needs to be retrained to stabilize and

depress the humeral head during the overhead motion. Just relieving pain without proper attention to the inhibited muscle group would only provide a temporary solution.

To simplify pain modulation for the purposes of this chapter, three manners of pain modulation will be presented. Sensory, rhythmical, and noxious mechanisms can explain the majority of pain modulation from common therapies (Table 32.2). Then, various treatments within each theory can be presented.

## TREATMENTS FOR PAIN

### Manual Therapy

The pain-modulating effects of most MT techniques has been suggested to be mainly neurophysiologic in origin, and may be mediated by the descending modulatory mechanisms (rhythmical and noxious). Current research suggests that this neurophysiologic response to MT is responsible for clinically significant decreases in pain.<sup>50</sup> It appears that different types of MT may work through different mechanisms (Table 32.3). Understanding the mechanisms may help clinicians choose which therapy is most appropriate for each patient, and may lead to more effective therapies in the future.

In a recent study investigating the effect of four MT techniques (manipulation/mobilization, muscle energy, massage, strain-counterstrain) performed on subjects with sites of muscle hypertonicity, tenderness, and joint restriction revealed that blood levels of  $\beta$ -endorphin and N-palmitoylethanolamide were elevated after treatment when compared to baseline. The greatest bio-marker alteration was experienced in subjects with chronic low back pain.<sup>51</sup>

**TABLE 32.2 Pain Modulation Mechanisms**

| Pain Modulation Theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Examples of Therapy                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sensory Pain Modulation<sup>47</sup></b><br>Sensory pain modulation describes Melzack and Wall's classic Gate Theory. Pain is propagated on smaller myelinated axons (A delta) and unmyelinated fibers (C fibers) and synapses in the spinal cord to ascend to the thalamus and then the sensory cortex where pain is processed. Stimulation of large diameter sensory fibers (light touch, vibration, etc.) in the painful area can inhibit the transmission of pain sensation at the spinal cord. Thus mechanical stimulation of the skin and surrounding tissues can "close the gate" at the spinal cord, minimizing the pain signal as long as the stimulation continues.                                                                                                                         | <ul style="list-style-type: none"> <li>• TENS</li> <li>• Massage</li> <li>• Intermittent compression</li> </ul>                                                                                          |
| <b>Rhythmical Pain Modulation<sup>48</sup></b><br>Rhythmical pain modulation is associated with a descending mechanism that elicits the production of endogenous opiates such as beta endorphin. This powerful analgesic method typically requires significant sensory input in a rhythmical fashion. Runner's high is one example of this type of pain modulation. The mechanical stimulation, which may include a muscle contraction, stimulates areas within the brain such as the Raphe nucleus and reticular formation to send signals to the pituitary to release beta lipotropin and adrenocorticotrophic hormone (ACTH). The beta lipotropin will form endorphin and the ACTH results in cortisol production at the adrenal glands. Each neurochemical can produce long-lasting pain modulation. | <ul style="list-style-type: none"> <li>• Rhythmical TENS</li> <li>• Myofascial release</li> <li>• Joint mobilization</li> <li>• Mobilization with movement</li> <li>• Muscle energy technique</li> </ul> |
| <b>Noxious Pain Modulation<sup>49</sup></b><br>Noxious pain modulation is the same as "central biasing" and descending noxious inhibitory control. This type of therapy requires a painful stimulus (C fiber stimulation) that elicits a cascade of neural factors in the midbrain and spinal cord to minimize pain. The periaqueductal gray is primarily responsible for producing enkephalins and serotonin when stimulated in this manner. This method can be combined with a rhythmical pattern for an added benefit.                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>• Instrumented soft tissue mobilization</li> <li>• Ice</li> <li>• Deep tissue massage</li> <li>• Trigger point therapy</li> </ul>                                 |

Modified from Denegar CR, Saliba E, Saliba S. *Therapeutic Modalities for Musculoskeletal Injuries*. 4th ed. Champaign: Human Kinetics; 2015.<sup>46</sup> TENS, Transcutaneous electrical nerve stimulation.

TABLE 32.3 Manual Therapy Application Guidelines

|                            |                                                                                                                                                                                                                                                                    |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Myofascial release         | Gentle sustained elongating pressure applied to the fascia for a minimum of 90–180 s<br>May be repeated until no further progress has been made                                                                                                                    |
| Muscle energy              | Isometric contraction:<br>Take joint into resistance (barrier)<br>Have patient contract 20%–50%<br>Hold for 7–10 s<br>Fully relax muscle<br>Take to next barrier, repeat until no further progress is being made (3–5 times)                                       |
| Mobilizations              | Depending on grade (3 and 4) the joint is placed at or into the first tissue stop<br>1–2 oscillations per second for 30–90 s<br>Reassess:<br>Take joint further into restriction<br>Repeat usually 3–5 times                                                       |
| Manipulation (HVLA)        | Patient is placed into the restricted movement until tension is felt<br>Patient should be relaxed<br>Practitioner imparts an HVLA into the restricted motion. (Cavitation may occur)                                                                               |
| Mobilization with movement | Usually performed weight bearing<br>Force directed in joint treatment plane<br>Should be minimal pain or pain free<br>Patient performs movement into the restriction while practitioner applies a force to help guide the motion<br>Repeat for 3–5 cycles of 5–10× |
| TPR                        | Find active TP:<br>Apply firm pressure to point<br>Stretch area so TP becomes more taught<br>Hold position until you feel the TP release (less taught and tender)<br>Stretch tissue further until TP becomes taught<br>Repeat process (usually 3–5 times)          |

HVLA, High velocity low amplitude; TPR, trigger point release.

Cryotherapy

Ice is the mantra for clinicians as the first line of defense for pain. Ice application is an effective analgesic and can control pain for several hours after an activity. Cooling the temperature of an inflamed area will also diminish the metabolic activity in the area, slow the production of inflammatory cells and potentially decrease the need for oxygen in the tissues. Following an acute injury, ice is combined with compression and elevation to help control swelling, and as mentioned previously, it should be combined with protection of the structure and controlled or optimal loading. This treatment can be used conveniently at home and can be administered several times per day.

Ice has been shown to be an effective modulator of pain in the treatment of soft tissue injuries.<sup>52–54</sup> Many clinicians believe that the cold application results in the spinal gate mechanism to diminish pain response from the focal area. However, the application of cryotherapy is typically noxious, with the patient perceiving cold, burning, aching, and then numbness. The mechanism for pain reduction is likely to be associated with a more complex neurologic pathway that stimulates smaller diameter nerve fibers, as described in the noxious pain modulation theory.<sup>54</sup> Ice has been shown to slow nerve conduction velocity and increase pain threshold.<sup>1</sup> Research evaluating pain relief is often confounded by increases in physical activity. Any positive change in activity should be interpreted as a reduction in discomfort that often limits function.

With the recent attention to elite athletes receiving treatment with cold baths or cold-water immersion following exercise, ice has been examined as a postrecovery modality. Again, few data support the use of cold therapy as a mechanism to prevent the inflammatory effects of exhaustive exercise or to minimize the effects on subsequent performance.<sup>55–58</sup> However, cold water immersion has been shown to help regulate core temperature when exercising in the heat.<sup>54</sup> Similar to other studies, methods to examine appropriate temperature, and the time and timing of the treatment are inconsistent, particularly when used as a recovery modality. Guidelines for temperature should be followed, since extreme cold temperatures have been associated with increasing the inflammatory process.<sup>59</sup> Temperatures below 50°F should not be used, despite the habituation to cold therapy, particularly with cold water immersion.<sup>60</sup>

There are numerous methods to apply ice or cryotherapy. An plastic bag filled with crushed ice is a convenient and inexpensive ice pack. Air insulates as an interface, so crushed ice is optimal, especially since cubes may be uncomfortable since they may increase pressure. Ice penetrates up to 4 cm, so it can be used to decrease the temperature of many superficial joints. However, adipose insulates against cooling, and longer treatment times are required depending on the treatment area. Icing a large muscle group, such as the quads, hamstrings, or hip, may require 40 minutes or more of treatment time, while the ankles and knees may require only 10 to 15 minutes of cooling for a therapeutic

effect. Generally, the more adipose present over the treatment site the longer the treatment required for adequate cooling.

Compression is often used in conjunction with ice and may be a confounder when investigating the therapeutic effects of cooling, since both may affect the development of edema.<sup>3</sup> Plastic wraps are often used to secure ice bags to the body so that sitting during the treatment is not necessary. Caution should be used with compressive wraps since a tourniquet effect can occur. Additionally, pressure with the cold can result in damage to superficial nerves, such as the ulnar nerve when treating the elbow, or the fibular nerve when treating the knee.

## Electrical Stimulation

Electrical stimulation is characteristically applied for pain relief. There are numerous waveforms with specific parameters that permit a clinician to adjust intensity (amplitude), phase duration, and pulse frequency, or there are clinical units that select combinations of parameters with the push of a button. Most units for therapeutic electrical stimulation can be classified as TENS units since they are applied superficially, passing a current through the skin to affect an intact nervous system. Subclassifications of stimulators are used to distinguish the goal of the treatment (either for pain relief or for facilitation of motor function) or the waveform of the stimulator (high voltage, biphasic, interferential, etc.). For the purposes of this chapter, sensory TENS units will be distinguished from neuromuscular stimulators (NMES) since the goal of the treatments are different. The sensory stimulators are primarily used for pain relief and are capable of eliciting a motor response, but the purpose is not to facilitate the muscle function. NMES units, conversely, are designed to stimulate a motor nerve to elicit a muscular response that may be used during rehabilitation to improve the ability to contract a specific muscle.

## Transcutaneous Electrical Nerve Stimulation

TENS operates by stimulating afferent nerves (sensory) to promote pain modulation. There are several proposed mechanisms for pain modulation, and the stimulation parameters associated with each mechanism are designed to enhance the effectiveness of that neuromodulation mechanism. For example, in the gate theory, large diameter sensory nerves are targeted to result in an inhibitory effect of the smaller pain fibers in the spinal cord. "Sensory" TENS is applied to target those large diameter nerve fibers to close the "gate" to modulate pain. Thus the parameters associated with "sensory" TENS are designed to maximize the stimulation of those large diameter nerves. Since the neuromodulation techniques are complicated, many modality companies have simplified the design of stimulators so the clinician can select the type of stimulation they want and the parameters will be packaged appropriately. However, in order for clinical trials to evaluate the usefulness of TENS, the parameter selection should be specified so that appropriate comparisons can be made, particularly with systematic reviews.

For the purposes of this chapter, the pain modulation mechanisms are categorized as (1) sensory TENS, (2) motor TENS, and (3) noxious TENS. These are common methods of applying electrical stimulation for pain relief and have application for

injury management. The sensory TENS mechanism, as previously mentioned, is based on the gate theory and inhibits pain in the spinal cord. It is often used with acute injuries since the stimulation is comfortable, but the analgesic effect is short lived. This treatment is appropriate for use prior to therapeutic exercise or after activity for pain management. The motor TENS mechanism is sometimes a misnomer, since an electrically produced muscle contraction is not required for this mechanism. Many times, there is a rhythmical muscle contraction and therefore it is not used as frequently for acute injuries. The neural mechanism is associated with a descending analgesic pathway, meaning that the pain is first perceived by the cortex. Then, specific brain centers, such as the pituitary gland, are signaled to produce precursors to the endogenous opioid, endorphin. Similarly, precursors are released that signal the adrenal glands to produce the hormone cortisol. This neural mechanism is elicited by rhythmical stimulation of A delta fibers, which are the larger of the pain fibers (but smaller than A beta fibers). Finally, noxious TENS, as its name implies, produces a strong, uncomfortable stimulation, similar to a bee sting. Theoretically, that type of stimulation targets the smallest pain fibers to alert specific brain centers to release enkephalin (another endogenous opiate), and serotonin, which both have powerful analgesic qualities. The parameter selection in the stimulator is based on the attempt to preferentially stimulate the target nerve in each of these pain modulation schemes.

## Edema/Inflammation

Interventions that may have an effect on edema and the inflammatory reaction include some ice, electrical stimulation methods, and bioenergies such as nonthermal ultrasound and laser. Mechanical methods of edema resolution include massage, compression, and any device that promotes muscle pumping, such as electrical stimulation. Ice traditionally has been thought to decrease blood flow to the cooled region, thereby diminishing the potential for edema formation. However, a recent study to determine the microperfusion effects of cold on skeletal muscle indicated that blood flow is not diminished at rest with therapeutic cooling.<sup>56</sup> Perhaps ice has the potential effect of attenuating the increased vascularity associated with an inflammatory event, although there is no strong evidence that ice alone reduces or minimizes edema.<sup>61-63</sup> Ice at extremely low temperatures has been shown to increase edema associated with rabbit tibial fractures, and caution should be used with low temperature ice or slush baths below 50°F.<sup>60</sup> Since ice has such strong effects on pain reduction, the treatment remains indicated as the standard of care in acute injuries.

Electrical stimulation has also been proposed for edema resolution. Although there is some evidence that high voltage stimulation can facilitate wound healing, it is difficult to ascertain whether high voltage can minimize or reduce soft tissue edema.<sup>64</sup>

## Ultrasound

Therapeutic ultrasound uses high-frequency acoustic energy to produce thermal and nonthermal (mechanical) effects in the tissues. The frequency of therapeutic ultrasound is clinically applied at either 1 MHz or 3 MHz with the higher frequency units directed to more superficial pathologies. Ultrasound is commonly prescribed

for pain and for tissue healing, or for the thermal benefits of increasing ROM or a reduction in muscle spasm. Similar to other modalities, the evidence for its use in improving outcomes in musculoskeletal pathologies remains equivocal.

## NONTHERMAL ULTRASOUND TREATMENTS

The nonthermal effects of ultrasound are most often associated with lower intensities and the pulsed mode of treatment. However, nonthermal effects occur with higher doses of ultrasound, but the effects are overshadowed by the thermal result. In pulsed ultrasound, the emission of acoustic energy is interrupted systematically to decrease the overall output and to allow time for thermal dissipation, which occurs with continuous wave application. Most commercial units have options of either 50% or 20% duty cycles, although many units are produced that allow the clinician more options.

The nonthermal effects of ultrasound, particularly pain relief and wound healing, are desirable in the acute phase of injury treatment since additional heating in an actively inflamed area may have detrimental results. The nonthermal effects, specifically cavitation and acoustical streaming are created by the mechanical pressure energy of ultrasound. Cavitation is the compression and expansion of gas bubbles that are present in the blood or tissues. This vibration causes mechanical and chemical alterations in the tissues. Although cavitation has been associated with negative tissue effects, it also has been associated with ion diffusion across cell membranes.<sup>65</sup> This process of applying low level mechanical energy to the tissues may affect fibroblastic activity in the inflammatory process, stimulate mitosis, and contribute to wound healing.<sup>66</sup> Acoustical microstreaming is associated with a “pro-inflammatory” effect. The acute processes, particularly the degranulation of mast cells, are accelerated slightly, which can result in a faster resolution of the initial phase of inflammation. This hypothesis has not been confirmed with controlled clinical trials, but pulsed ultrasound at low intensities may lead to faster resolution of injury when applied early.

### Stress Fracture Healing

Nonthermal ultrasound has been suggested to treat stress fractures. Low-intensity ultrasound (less than 3 mW) has been proposed as a treatment for nonunions and is currently FDA approved for these conditions. The application of either of these low-intensity mechanical or electromagnetic energies is proposed to stimulate the production of osteoblasts. Therefore there are bone stimulator units that deliver either ultrasound (acoustic) or electromagnetic (electrical field) energies, both of which have been used off-label for stress fracture management. Currently, there are few data on stress fracture healing to substantiate the use of these units. However, in the interest of trying any method of accelerating the healing phase for elite athletes, there are empirical observations of these treatments contributing to stress fracture management.

A specific device must be used to apply ultrasound for fracture healing, and the treatment cannot be reproduced on a standard therapeutic ultrasound machine, since the intensity is 1000 times lower. The units are designed for home treatment for nonunion

fracture healing; therefore the units are preprogrammed with the appropriate dosage. The device provides a stationary treatment for a set treatment time at a given intensity. Because the intensity is so low (approximately 0.003 W/cm<sup>2</sup>), there is little consequence to the stationary treatment. The treatment can be applied daily, and more frequent treatments are *not* suggested to improve outcomes. Therefore when these units are used for stress fracture healing, they are more likely to have the treatment applied in a clinic visit to prevent a coach or athlete from abusing the treatment. Empirically, these units have been used in conjunction with activity modification for stress reactions and stress fractures. One issue with conducting a randomized control trial is the subject's willingness to potentially receive the sham treatment. When the device is available, many runners decline the possibility of being randomized into the control condition.

### Phonophoresis

Cavitation has been associated with phonophoresis, which is the transfer of medications through intact skin using ultrasound. Phonophoresis—similar to iontophoresis (which uses electricity to transfer ions into the skin)—is used to administer medications locally to theoretically maximize the effect at the injured area. There are several conflicting studies that have investigated phonophoresis using lidocaine or antiinflammatory drugs, including steroids in clinical trials.<sup>46,67-69</sup> The cavitation of ultrasound may change the structure of the stratum corneum, which is the barrier to transdermal delivery. Although phonophoresis increases the penetration of many medications through the skin, there are no specific guidelines on the best clinical method. Furthermore, some medications may block the transmission of ultrasound.<sup>70</sup> The variety of parameter choice and application method in clinical practice may explain the inconsistent research results. The best practice methods indicate the use of a drug that has good ultrasound transmission must be applied, and lower frequencies are more optimal for phonophoresis. The lower frequencies are recommended despite evidence that the higher frequencies concentrate the energy at the skin. Drugs should also be at a consistent concentration (not diluted with ultrasound gel) and may need to be in place longer than the typical 5-minute treatment.<sup>71</sup> The suggested clinical approach would be to apply the drug directly to the skin and apply an occlusive or hydrogel dressing to hydrate the skin and prevent dilution of the drug. A lower frequency of ultrasound should be used while the remaining parameters of ultrasound are based on the energy dosage desired. Pulsed ultrasound with a low intensity is used for more acute pathologies while continuous ultrasound and higher intensities are used for subacute conditions. Longer treatment times with lower doses of ultrasound may not affect drug absorption; therefore the drug absorption is not necessarily associated with thermal effects of ultrasound.<sup>72</sup> Drugs that have shown beneficial effects with phonophoresis are lidocaine, dexamethasone, and ketoprofen, while hydrocortisone has not produced positive effects (Fig. 32.3).

### Iontophoresis

Iontophoresis is the transdermal administration of a medication with electrical current. The solution is placed in the active

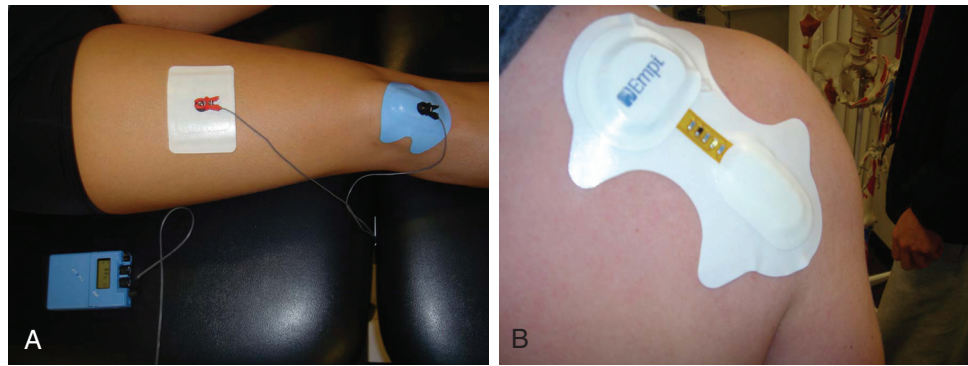


Fig. 32.3 Iontophoresis.

electrode and applied over the injured area, which is generally a focused point such as a tendon, bursa, or ligament. Iontophoresis requires galvanic or DC current, a monophasic waveform with an unlimited phase duration. The current “drives” ions of the same charge as the active electrode; therefore it is imperative that the polarity of the medication is known. This electrical propulsion is from like charges repelling; therefore a negatively charged ion should be placed in the negative electrode and a positively charged ion should be placed in the positive electrode. A second electrode (the dispersive) is placed at another location to complete the circuit. Similar to phonophoresis, the technique theoretically improves the transmission of a medication to a particular location and minimizes the effects of oral medication or undesired effects of an injection. Although there is evidence that iontophoresis increases the absorption of specific medications, clinical trials have not conclusively demonstrated an adequate effect using this technique.

There are many drugs that have been applied with iontophoresis (e.g., lidocaine), many corticosteroids, and botulinum for hyperhidrosis. NSAIDs, such as diclofenac and ketorolac, also have been used.<sup>73</sup> There are commercial devices that are marketed specifically for iontophoresis, and those units can be programmed to deliver a set amount of current over time (milliamp minutes). The current should be set to patient comfort and the duration of the treatment is automatically adjusted so that the total dosage, either 40 or 80 mA/min is applied. Shorter durations with higher amplitudes are not recommended since this does not affect the drug absorption.<sup>74</sup> State practice Acts may regulate the use of even topically administered medications by physical therapists or athletic trainers, so clinicians should be aware of the local regulations prior to use.

## Laser

Light therapy has been used for centuries and laser is concentrated light at a specific wavelength. Laser is an acronym for Light Amplification of Stimulated Emission of Radiation and was invented in the 1960s. Now, lasers are used in industry, entertainment, medicine, and all aspects of technology. Laser therapy, often called low-level laser therapy (LLLT) has recently been approved by the US Food and Drug Administration for the treatment of some musculoskeletal conditions such as carpal tunnel or cervical spine pain. There are several types of LLLT

available for clinicians as well as many parameters to be manipulated to achieve the desired treatment outcome. More research is needed to expand the capabilities of this modality in providing potential benefits to the patient.

Therapeutic lasers are often difficult to discriminate between liquid crystal diodes (LCDs) light and it is important to read the product literature to understand the total power of the laser being used. Many companies combine a single laser with multiple LCDs, and the incorrect implication may be that all the lights are emitting similar energy. Lasers have unique properties—a single wavelength—and are able to penetrate into the skin and tissues better than other types of light. The biological effects of the laser are associated with the absorption of energy and the resultant production of ATP, an energy source to a cell. The cells respond by accelerating their normal activity and can thus facilitate the inflammatory response. LLLT has also been used effectively for pain; however, the mechanism for pain modulation is unknown.

Parameter selection is perhaps the most difficult aspect of LLLT since there are so many variables that must be considered.<sup>75</sup> There are few tested guidelines of dosing for specific pathologic entities and determining the amount of energy to be applied is often vague. Patient considerations include the type and condition of the tissue, and acute or chronic condition; even pigmentation may have an impact on the laser treatment outcome. Laser parameters that must be considered include wavelength, output power, average power, power density (intensity), and energy density (dosage). Many of the LLLT therapy devices marketed are menu driven, so selecting the tissue type and chronicity of the injury provides a specific treatment dosage. The total energy often is clinically determined by the treatment time. For example, once the unit has been purchased, the wavelength, power, and density are set, and the clinician determines the dosage by increasing or decreasing the treatment time. Each area should be treated for that time period, so units with more than one laser diode are more efficient, particularly when treating larger areas.

One of the problems with evaluating the evidence of LLLT is that laser treatment is often poorly documented in the literature. Different laser types were used and dosages were varied or not reported. Laser research should report the methodology and describe all factors of the clinical unit so that outcome-based evidence can be evaluated.

LLLT has been proposed for many conditions, such as pain and wound healing, and inflammatory conditions such as bursitis, tenosynovitis, or acute musculoskeletal traumas. Since LLLT is only currently approved for specific pathologies, an off-label treatment would be necessary. Higher energies are often more desirable, since many inconclusive studies used very low energy levels.<sup>76,77</sup> The treatment time to deliver an effective treatment increases the total amount of energy delivered. Treatment doses range from 4 to 50 J and it has been suggested that no more than 50 J should be applied to adults and no more than 25 J should be applied to children under the age of 14.<sup>75</sup>

MT is often used to mechanically affect edema. Once the injury has stabilized and there is hemostasis, light massage in the direction of lymphatic flow can facilitate the resolution of edema. Deeper massage and soft tissue techniques can be used to mobilize more brawny or pitting edema.

### Compression: Intermittent Compression

In an attempt to provide this mechanical stimulation to reduce edema, devices have been manufactured to apply compression to the extremities. Various agents use intermittent compression therapy and newer units have sequential pressure chambers to increase pressure in the extremities from distal to proximal. The mechanical pressure provides a sensory stimulation, which often can reduce pain or facilitate recovery after exercise.<sup>78</sup> Most of the evidence to support the intermittent compression devices and tissue oscillation therapy are empirically based (Fig. 32.4).

### Range of Motion

Improving ROM is an important therapeutic goal and attention should be made to have the patient actively use or move the joint after efforts to improve the range. When passive techniques, such as superficial heat, thermal ultrasound, or joint or soft tissue mobilization, are combined with active exercise, the patient learns to contract the muscle in the new range. Muscle inhibition often accompanies joint or soft-tissue tightness. Although “warming up” the area dynamically or with a passive intervention prepares the

site, it is essential to do muscle re-education techniques directly following the initial treatment.

### Superficial Heat

Superficial heat in the form of a moist heating pad may be used to help warm the body part up or to prepare for exercise. Heat should not be used if there is swelling or pain, but it can be very effective several days after a significant hamstring strain, for example, when there is ecchymosis and blood accumulated in the tissues. The increased vascularity helps mobilize and breakdown the fibrous hematoma. Like any other modality, the application of the agent in isolation does little to change the state of the condition. Therefore superficial heat, even with a hot tub or warm shower, should be followed closely with stretching, focused exercise to address the problem, or with MT techniques.

When superficial heat is applied, skin receptors detect the increased temperature, and there is a vasodilatory response in the skin and subcutaneous tissues. This protective mechanism helps to preserve core temperature and to dissipate the local increased temperature to prevent burns. If there is a vascular compromise, heat should not be used, and the addition of compression may prevent the appropriate dissipation of heat. Thus sitting on hot packs is not recommended since this might result in a local ischemic response associated with pressure. Because of the vascular response of the skin, there is very poor transmission of superficial heat deep into muscle. Intramuscular thermistors show only a 1 cm penetration that is affected by the superficial adipose layer.<sup>79</sup> Therefore if the goal is to provide heat and increased vascularity to a muscle, active exercise is the best choice rather than a hot pack. Superficial heat will penetrate many joints, such as the knee, wrist, or ankle, and may be effective prior to joint mobilization. Deeper tissues such as the hip, facet joints of the spine, and perhaps the shoulder may not respond, except through a cutaneous mechanism. There is some evidence that superficial heat may elicit a reflex action to underlying musculature, despite the lack of temperature increase in the muscle.

Heat often has a relaxing, sedating effect; therefore it is commonly used with muscle spasm.<sup>80</sup> Caution should be used because of a commonly documented phenomenon called heat hyperreflexia. The patient uses heat to relax the tissues and generally has good results during the application. Then, several minutes or hours after the treatment, there is an increase in pain, which is often associated with an exacerbation of muscle spasm. Focused heating with directed exercise while the tissues are heated may prevent this occurrence; however, superficial heat should be used judiciously with overuse injuries. Many times, spinal injuries are treated with heat, even in the acute phase, to promote relaxation. However, the evaluation should determine the phase of inflammation and the amount of activity that also results in heating of the tissues. Generally, if the condition is “tight,” then heat can be used effectively. However, if the ROM is limited by “pain,” then cold may improve the outcome.

Heat can theoretically affect the compliance of connective tissue by influencing the viscoelastic ground substance. Once the tissues are heated to an effective temperature of 40°C (104°F), there is less stress needed to change tissue length. Therefore if the goal is to improve ROM or joint mobility, heating prior to

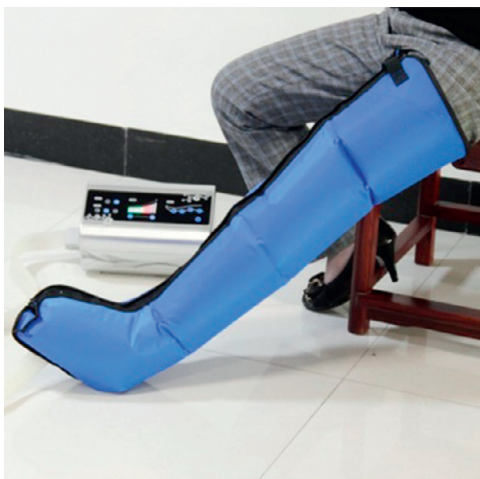


Fig. 32.4 Intermittent compression.

stretching or joint mobilization techniques may be effective. Higher tissue temperatures result in the denaturation of proteins, and burns result. Because of the limitation of superficial heating agents in increasing tissue temperatures beyond the skin, deep heating agents are often used. Stretching or joint mobility techniques should be done while the tissues are heated.<sup>79,81</sup> Simultaneous heat and stretch may accomplish this goal. Patient positioning with the appropriate tissues heated are essential in this technique. A common mistake is to place a hotpack over the top of a shoulder, for example, when the tight structures are generally in the axilla. Thus far, data that demonstrate a clinical benefit of using superficial heat on the ROM are sparse.<sup>82,83</sup> Furthermore, if increasing muscle temperature is the goal, active exercise is the best method. Activity improves blood flow and intramuscular temperature more effectively than hot packs or deep heating agents such as ultrasound.

Superficial heat has many therapeutic forms. For a home program, a moist towel can be heated in the microwave and applied to the body. The towel can be placed in an open zip-lock bag. Extreme caution should be used since burns are likely with this technique. Dry towels should cover the area completely. Likewise, cooked rice or a boiled egg holds heat for 20 to 30 minutes and can be used to apply heat to a focal area. Hot tubs, baths, and heating pads also can be used, but the patient should be directed to use short durations and then stretch or perform their stabilization exercises immediately. Hot tub temperatures should not exceed 100°F, although the extremities can tolerate temperatures of 104 to 107°F. Hot tubs potentially stress the thermoregulatory effects and should not be used after exercise.

## THERMAL EFFECTS OF ULTRASOUND

Therapeutic ultrasound is most commonly applied as a deep heating modality. The acoustic energy penetrates well through superficial tissues, such as adipose, and the mechanical, vibrational energy is converted to thermal energy when absorbed by the tissues. Tissues that absorb ultrasound energy are those with high protein content, such as muscle, tendon, and nerve. Ultrasound does not penetrate through bone; therefore the energy reflects off bone and potentially magnifies the effect at the bone/muscle interface. Because the temperature changes occur in tissues other than the skin where there are no thermal receptors, there is little perception of heating other than mild warmth associated with the treatment. Likewise, since the energy is selectively absorbed, reflected, and refracted, the recommended doses of ultrasound are difficult to determine. Patients, particularly athletes, often request ultrasound treatments and since there is little perception of energy emitted, it is difficult to examine the effects of the treatment. However, tissue temperatures can increase substantially, and energy is selectively absorbed in the muscle, tendon, and nerve, and periosteal burns may result from higher doses. Indiscriminate use or self-prescribed treatments should never be condoned.

Ultrasound dosage is specified by the frequency, intensity, duration, and duty cycle. As stated previously, the lower intensities and lower duty cycles reduce the possibility of a thermal effect and are often prescribed for nonthermal mechanisms. A lack of a duty cycle, or continuous wave ultrasound (100%) is

generally used for thermal effects. The frequency and intensity are inter-related, although each has specific effects. The frequency affects the depth of penetration, with higher frequencies concentrating the energy more superficially. A 3 MHz ultrasound will generally have a depth of penetration of approximately 1 to 3 cm, while a 1 MHz ultrasound will have a depth of penetration of 3 to 8 cm. However, the depth depends on the absorption of energy within the tissues. For example, if the energy is preferentially absorbed by tissues with a high protein content, such as tendon, then there is less energy to transmit; thus the depth of penetration is affected. Generally, when applying ultrasound for thermal effects in deep tissues such as a muscle belly, the lower frequencies are used, and when there is superficial bone the higher frequency is used. Thermal energy can also transmit via conduction, so the science in determining a specific frequency and intensity for a specific thermal effect is inexact.

Thermal ultrasound is commonly used to help break down scar tissue, to promote the alignment of collagen tissue in the remodeling phase, and to increase ROM when combined with stretching or mobilization techniques. Currently, there are no data to support the use of thermal ultrasound to affect blood flow to tissues other than subcutaneous structures. When attempting to increase ROM, it is important to stretch the connective tissue either during the ultrasound treatment or immediately afterward while the temperature remains elevated. Ultrasound treatments without the concomitant therapeutic exercise or MT technique have poor results.

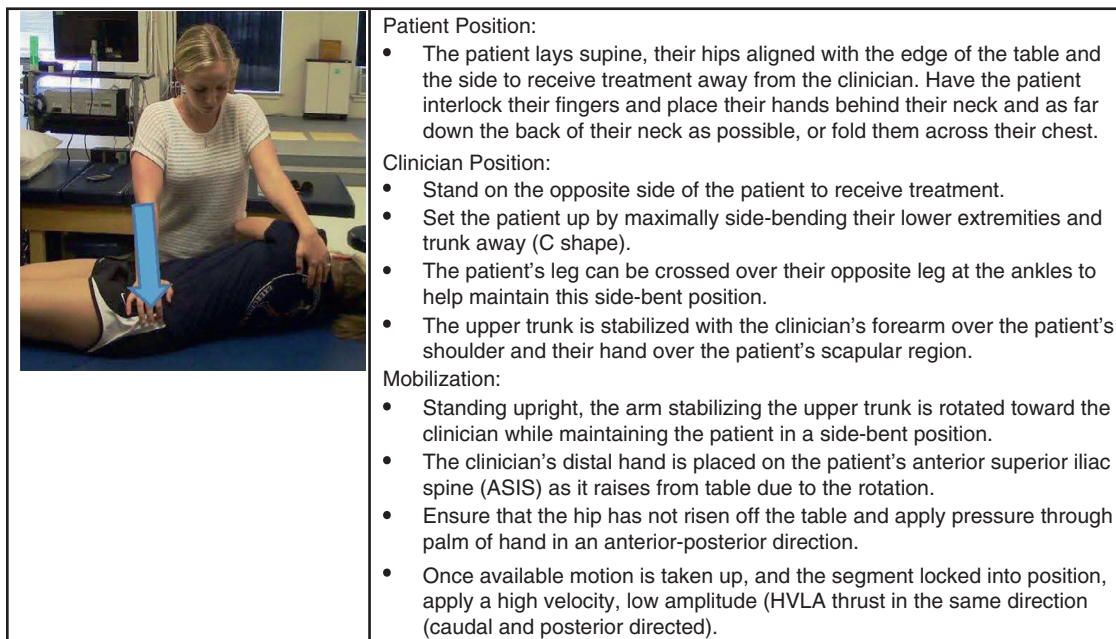
## MOBILIZATION/MANIPULATION

An MT technique comprising a continuum of skilled passive movements to the joints and/or soft tissues is applied at varying speeds and amplitudes, including a small amplitude/high velocity therapeutic movement.<sup>2</sup> Research has shown that mobilization used as a therapy can produce significant mechanical and neurophysiologic effects.<sup>50,84,85</sup> The explanations of these effects—the mechanism of mobilization—is still relatively unknown, especially in regards to the spine, and is subject to further research.<sup>86</sup> However, several theories have been established in accordance with effects seen, including the effects of pain relief, increasing ROM and the influence on the autonomic nervous system.

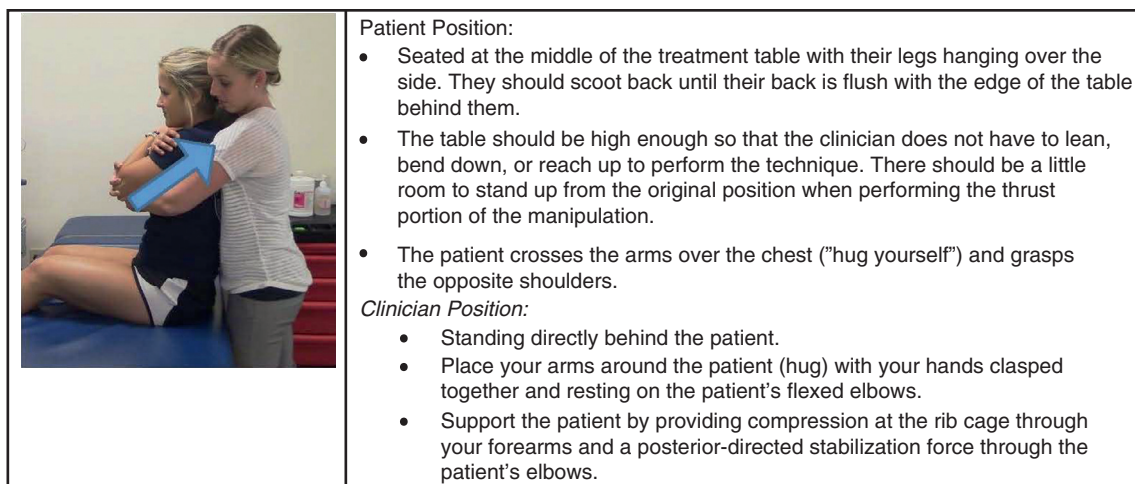
Multiple reviews have investigated the pain-modulating mechanisms of spinal manipulation<sup>87,88</sup> and agree that the analgesic origins are neurophysiologic in nature, occurring through some type of descending pain modulation circuit. However, the exact circuit is not fully understood, but is becoming clearer through the work of Bialosky et al.<sup>50</sup> and it appears that different types of spinal manipulations—namely the velocity with which and the location at which they are performed—may elicit different neurochemical responses indicative of different descending pain modulation mechanisms. Please refer to Figs. 32.5 to 32.9 for examples of common mobilization/manipulation techniques.

## MOBILIZATIONS WITH MOVEMENT

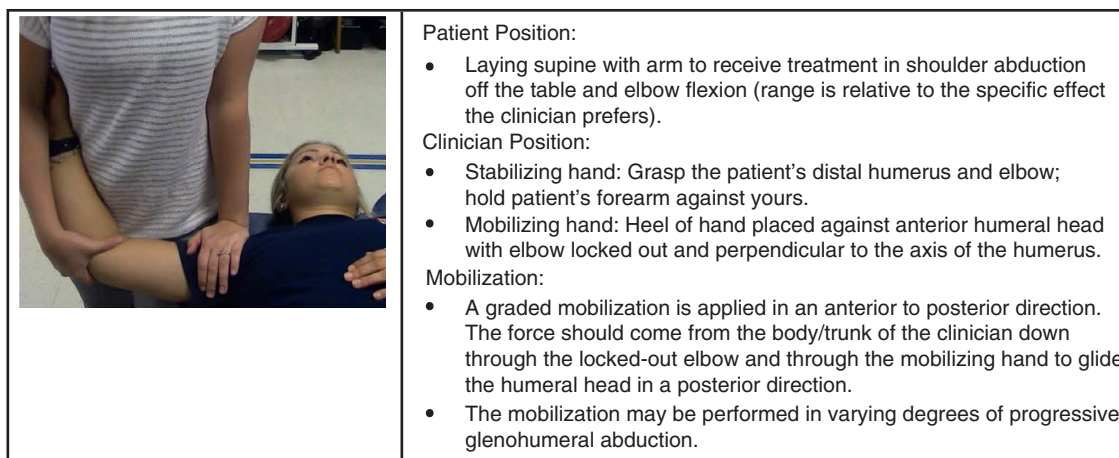
The concept of mobilizations with movement (MWM) of the extremities and sustained natural apophyseal glides (SNAGS)



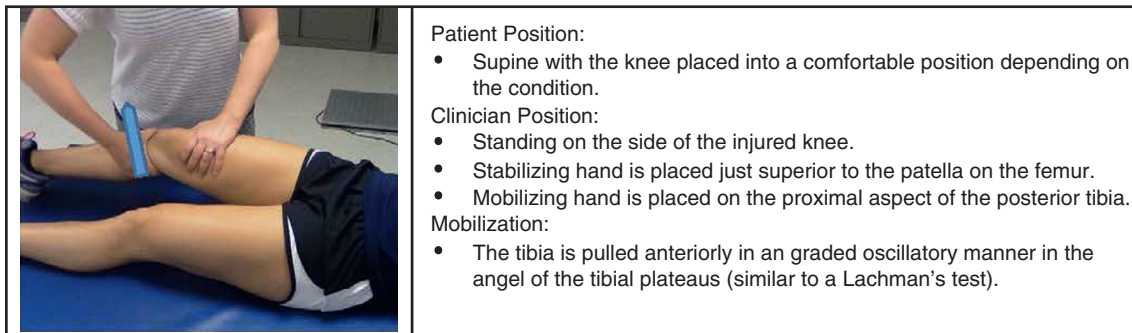
**Fig. 32.5** Lumbopelvic manipulation. *HVLA*, High velocity, low amplitude.



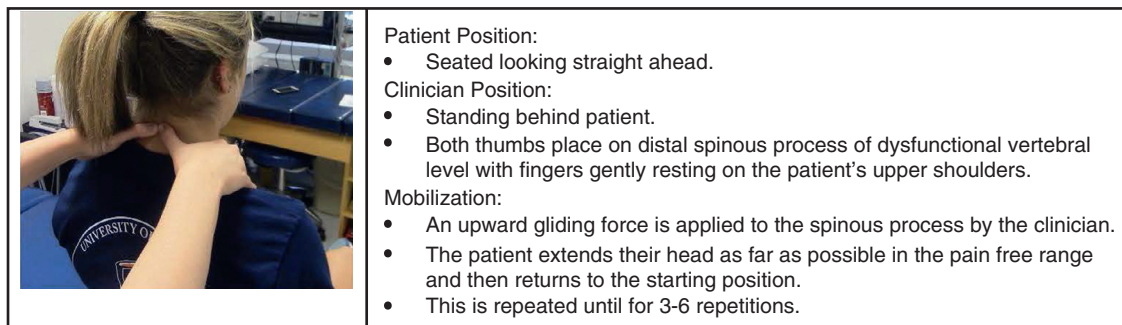
**Fig. 32.6** Mid-thoracic manipulation.



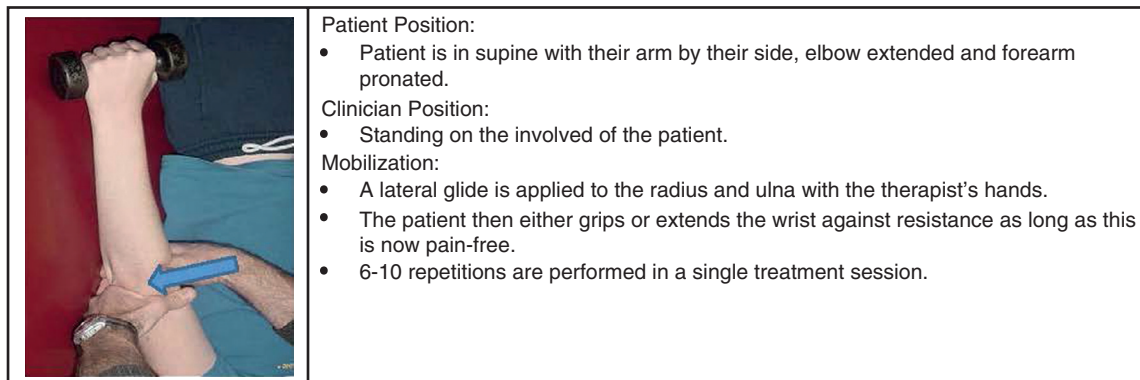
**Fig. 32.7** Superior-inferior glenohumeral mobilization (into joint resistance).



**Fig. 32.8** Tibiofemoral posterior-anterior (mobilization).



**Fig. 32.9** Extension SNAG (cervical spine). Indications: facet joint dysfunction, cervical pain with extension, SNAG, Sustained natural apophyseal glides.



**Fig. 32.10** Mobilization with movement for lateral epicondylalgia.

of the spine were first coined by Brian R. Mulligan.<sup>89</sup> MWM is the concurrent application of sustained accessory mobilization applied by a therapist and an active physiologic movement to end range applied by the patient. Passive end-of-range overpressure, or stretching, is delivered without pain as a barrier.<sup>89</sup> Research has shown that MWM can be beneficial from increasing ROM and decreasing pain; however, the mechanisms by which it affects the musculoskeletal system, whether mechanical or physiologic, has not been fully determined. Common MWM for the spine are SNAGS (Fig. 32.10), which have been shown to decrease pain, disability, and headache; increase ROM and patient satisfaction; and be as effective as manipulation and other forms of mobilization. However, there is no evidence that these techniques resolve the biomechanics of the facet joint.<sup>90,91</sup> Several placebo-controlled studies<sup>92,93</sup> and a case series<sup>94</sup> have demonstrated that

a single application of the MWM for upper extremity results in an immediate increase in pain-free grip force (strength).

## MUSCLE DYSFUNCTION

Muscle dysfunction commonly accompanies orthopaedic injury, and stiffness, spasm, and muscle tightness may create more symptoms than a joint pathology. Articular receptors create neuroreflexive responses, making muscle inhibition (arthrogenic muscle inhibition), reducing muscle activation, and triggering points a consequence of joint injury. Interventions designed to reduce muscle inhibition, to reduce tension and spasm, and to improve timing and activation are appropriate therapies. Various treatments, such as cryotherapy on the joint, NMES and PENS, MET, PNF, and trigger point therapy (TrPT) are effective to

change muscle function. Similar to all passive treatments, responses to the therapy should be measured, and therapeutic exercise should be initiated following the intervention.

## JOINT CRYOTHERAPY TO COMBAT MUSCLE INHIBITION

The studies on muscle activation describe the usefulness of combining modality treatment with rehabilitation, rather than passively applying the treatment.<sup>95</sup> Ice is applied in time intervals where the joint is cooled, then exercise is conducted within the pain limits and the joint is re-cooled. This procedure is known as cryokinetics and is used to promote joint mobility and muscle function in the acute stage of healing.<sup>96,97</sup> Ice should be applied in the acute phase several times a day, if possible. However, there are currently no guidelines on the number of times per day, how long each treatment should last, or how frequent the treatments should be, based on clinical trials or outcome studies. Therefore treatments should be prescribed with pain as a guide for acute injuries. For example, during the early phase when pain is a problem, more frequent treatments are necessary. However, as the acute pain resolves, ice treatments should be used before and after therapeutic exercise to facilitate the muscle and to monitor postexercise inflammation.

Cryotherapy has been receiving attention as a complement to therapeutic exercise with its effect on muscle activation. Arthrogenic muscle inhibition (AMI) is a condition that often follows joint injury where the muscles surrounding that joint are reflexively inhibited. AMI exists despite rehabilitation efforts, and may result in abhorrent forces through a joint over time.<sup>98</sup> Focal application of cryotherapy over a joint has been shown to decrease AMI.<sup>99</sup> Using ice before therapeutic exercise as a mechanism to improve muscle activation could improve the outcomes of rehabilitation.

## NEUROMUSCULAR ELECTRICAL STIMULATION

A different approach to electrical stimulation is to use it to stimulate the motor nerve to improve or facilitate muscle contractions. Since the treatment is directed at the motor nerve ( $\alpha$ -motoneuron), the treatment is labeled neuromuscular electrical stimulation or NMES. This treatment should not be confused with Motor TENS, which is for pain relief. NMES is used to retrain a muscle to function, to overcome arthrogenic inhibition, or to permit exercise when there is an intact motor nerve. NMES can be used to improve the strength and coordination of motor units when used in conjunction with voluntary exercise, but the device has not been shown to strengthen or reduce adipose with passive use. The treatment has been shown to improve some outcomes in anterior cruciate ligament rehabilitation when used in conjunction with exercise therapy.<sup>100</sup>

NMES can be used when a specific muscle cannot be exercised volitionally. An effusion can result in a reflex inhibition of muscles that surround a joint and it is difficult to elicit a good contraction. The electrodes are placed over the muscle, generally with one electrode over the superficial aspect of the nerve to the muscle and one electrode over the motor point. The phase

duration should be between 200 and 400  $\mu$ s, on the higher end of the TENS range because the targeted nerve is relatively deep. The amplitude should increase so that a tolerable and visible contraction is noted. The pulse rate should be at tetany, which is 35 to 50 pps.

Clinically, NMES is commonly used for quadriceps contractions and retraining following a knee injury or surgery. It is also effective in promoting lower trapezius function during shoulder and scapular exercise. The treatment also can be used to help retrain muscle function after a tendon transfer or following a nerve injury when the muscle is innervated, but the function is limited. Again, fatigue is common and overworking those muscles can further deteriorate function. Generally, once the ability to exercise the muscle volitionally has returned, the usefulness of this modality is limited, since active exercise is preferred.

## PATTERNED ELECTRICAL NERVE STIMULATION

PENS is a different form of electrical stimulation based on electromyographic activity during functional tasks. The purpose of PENS is to replicate the normal firing pattern through a muscle re-education technique that delivers precisely timed stimulation of both the agonist and antagonist muscle groups. PENS has been applied to a variety of populations with improved results including knee osteoarthritis<sup>101</sup> and patellofemoral pain (Fig. 32.11).<sup>25,102</sup>

## MUSCLE ENERGY TECHNIQUE

MET is a MT and involves the voluntary contraction of a muscle in a precisely controlled direction at varying levels of intensity against a distinct counterforce applied by the operator.<sup>103</sup> The five basic elements of MET are patient-active muscle contraction, controlled joint position, muscle contraction in a specific

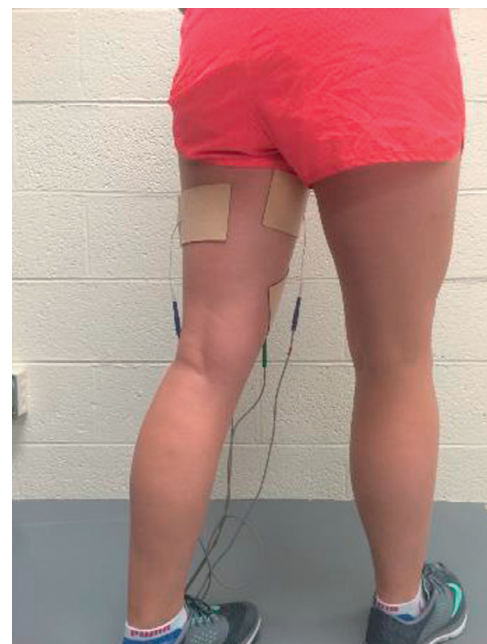
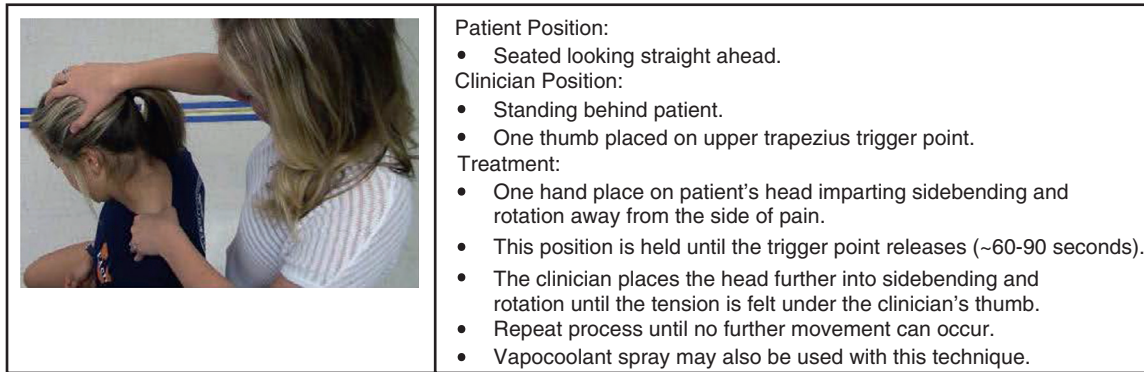


Fig. 32.11 Patterned electrical nerve stimulation.



**Fig. 32.12** Trigger point hold and release for upper trapezius.

direction, operator-applied counterforce, and controlled contraction intensity. Postisometric relaxation (autogenic inhibition) and reciprocal inhibition are the two neurophysiologic principles on which MET is based. There are three contractions that are utilized in MET: isometric, isotonic, and isolytic (eccentric). MET has been shown to decrease pain<sup>104</sup> and segmental dysfunction by increasing the pain threshold and increasing joint ROM and muscle mobility with effects that can be last for 1 week.<sup>105</sup>

## MYOFASCIAL RELEASE/TRIGGER POINT THERAPY

These MT techniques are being discussed together because of the similarities with signs and symptoms and treatment, but they can be divided into two separate and distinct therapies. Myofascial release and TrPT are specialized MT techniques that attempt to effectively release fascial tension and restrictions. Fascia can become dysfunctional through any type of stressor (e.g., trauma, illness, postural dysfunction, psychological stress).

When fascia becomes problematic it causes pain, tenderness, and trigger points in local tissues. Trigger points are defined as areas of tenderness and hyperirritation, exhibition taut bands, proprioceptive changes, and a consistent pain referral pattern.<sup>106</sup> An example of TrPT for an upper trapezius trigger point is shown in Fig. 32.12.

## CONCLUSION

There are numerous devices in the arsenal to facilitate rehabilitation techniques, particularly in a patient population with expectations of high activity levels. However, it is important to understand the limitations of physical agents and MT and to put their use in context with specific therapeutic goals. Physical agents can diminish the inflammatory response and are often effective in managing pain, but modality use in isolation is not recommended. The combination of therapeutic modalities and MT gives the clinician the best chance for successful outcomes.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

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